

PROJECT 2 - STREAMLINING IT PROCUREMENT: AUTOMATING STANDARD LAPTOP ORDERS WITH FLOW DESIGNER

ABSTRACT

IT procurement is the process by which an organization requests, approves, buys and prepares hardware such as laptops for its employees. When this process depends on manual follow-up, the configuration of a newly arrived laptop is easily forgotten or delayed, which increases user waiting time and wastes the effort of the IT department. This project, "**Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer**", removes that manual step by using the low-code **Flow Designer** of the ServiceNow platform.

A flow named "**Standard Laptop task**" was created in the Global application with the run-as user set to System User. It is triggered by the **Service Catalog** trigger and uses the **Create Catalog Task** action to generate a catalog task (sc_task) for the requested item automatically. The task carries the short description "Laptop needs to Configured", the assignment group **Hardware** and the approval value **Approved**. The flow was then linked to the **Standard Laptop** catalog item through the Process Engine tab of the item.

The solution was tested by ordering the Standard Laptop (Lenovo Carbon X1) from the Service Catalog. The request REQ0010001 was approved, and a catalog task, SCTASK0010001, was created automatically under the requested item RITM0010001 and assigned to the Hardware group. The project shows that workflow automation reduces manual intervention and errors, shortens the time to configure laptops, and improves the use of IT staff.

Keywords: ServiceNow, Flow Designer, Service Catalog, Catalog Task, IT Procurement, Workflow Automation, Low-Code.

1. INTRODUCTION

1.1 Overview

Every organization needs to supply its employees with laptops and other equipment. In a typical IT department, a request is raised, approved, ordered from a vendor and, once the device arrives, configured with the required operating system, software and security settings. Each hand-over between people is a chance for delay: a task may not be created, may be sent to the wrong team, or may be forgotten.

ServiceNow is a cloud platform for enterprise workflows. Its **Flow Designer** is a low-code tool that lets developers and administrators build automations from triggers, actions and flow logic without writing large amounts of code. In this project, Flow Designer is used to automatically create a configuration task for the hardware team as soon as a standard laptop request is placed through the Service Catalog.

1.2 About the Naan Mudhalvan Programme

Naan Mudhalvan is a flagship skilling initiative of the Government of Tamil Nadu, implemented by the Tamil Nadu Skill Development Corporation (TNSkills). It provides students of arts, science and engineering colleges with industry-aligned training and certification in emerging technologies so that they become employable. This project has been carried out as part of the ServiceNow course offered under the programme.

1.3 Problem Statement

The current IT procurement process lacks efficiency and automation, resulting in delays and manual overhead, particularly in handling standard laptop orders. Requests for standard laptops often require configuration, but this step is prone to oversight or delay, leading to frustration among users and inefficient resource allocation within the IT department.

1.4 User Story

As a member of the IT procurement team, I want a streamlined process for ordering standard laptops so that tasks are automatically generated and assigned to the hardware team for configuration. This will ensure that laptops are promptly configured upon arrival, reducing user wait times and minimising manual intervention.

1.5 Objectives

1. Create a seamless experience for users requesting standard laptops by ensuring timely configuration.
2. Reduce manual intervention and potential errors in the procurement process.
3. Improve resource utilisation within the IT department by optimising task allocation.
4. Enhance overall efficiency and productivity in IT procurement operations.

1.6 Scope of the Project

The project covers one catalog item, **Standard Laptop**, in the Global application scope. It includes creating and activating a flow, linking it to the catalog item, and testing the complete cycle from ordering to task creation. Purchase-order integration with vendors, asset registration and e-mail notifications are outside the scope of this work.

2. SYSTEM STUDY

2.1 Existing System

- A laptop request is approved, but the follow-up task for configuration is created manually by a member of the IT team.
- The person responsible may forget or delay the step, so the laptop waits unconfigured after it arrives.
- Tasks may be assigned to the wrong team or to no team, which wastes effort and time.
- Users have no clear view of the progress of their request and become frustrated.

2.2 Proposed System

In the proposed system, a flow runs automatically whenever a Standard Laptop is ordered from the Service Catalog. The flow creates a catalog task for the requested item, gives it a clear short description, and assigns it to the **Hardware** group. The main features are:

- Automatic creation of a catalog task for every standard laptop order.
- A standard short description and description, "Laptop needs to Configured", so the hardware team knows the action required.
- Automatic assignment to the Hardware group, which removes manual routing.
- The flow is linked to the catalog item, so no separate action is needed by the procurement team.

2.3 Feasibility Study

- **Technical feasibility:** Flow Designer, the Service Catalog and catalog tasks are standard ServiceNow features and need no external tools.
- **Economic feasibility:** The solution uses the existing platform and only configuration effort, so the cost is low; it saves time in every order.
- **Operational feasibility:** The hardware team continues to work with normal catalog tasks, so little training is needed.

3. SYSTEM REQUIREMENTS

3.1 Hardware Requirements

Table 3.1: Hardware requirements

Component	Specification
Processor	Intel Core i3 or above (or equivalent)
RAM	4 GB minimum (8 GB recommended)
Storage	256 GB or above
Network	Stable internet connection

3.2 Software Requirements

Table 3.2: Software requirements

Component	Specification
Platform	ServiceNow Personal Developer Instance (PDI) / training instance
Access level	Administrator (admin)
Tool used	Flow Designer (Process Automation)
Web browser	Google Chrome / Microsoft Edge (latest version)
Application scope	Global

3.3 ServiceNow Components Used

Table 3.3: ServiceNow components and their role

Component	Table	Role in the project
Catalog Item - Standard Laptop	sc_cat_item	The item ordered by the user; linked to the flow
Request	sc_request	Created when the order is placed (REQ0010001)
Requested Item	sc_req_item	The ordered laptop inside the request (RITM0010001)
Catalog Task	sc_task	Task created by the flow for the Hardware team (SCTASK0010001)
Approval	sysapproval_approver	Approver decision on the request
Flow - Standard Laptop task	sys_hub_flow	The automation that creates the catalog task

4. SYSTEM DESIGN

4.1 Workflow Overview

The end-to-end process combines the manual actions of the requester and approver with the automatic actions of the flow.

Table 4.1: End-to-end process

Step	Activity	Performed by	Result
1	Order the Standard Laptop from the Service Catalog	Requester	Request REQ0010001 and requested item RITM0010001 are created
2	Service Catalog trigger fires	Flow (automatic)	The flow "Standard Laptop task" starts
3	Create Catalog Task action runs	Flow (automatic)	Task SCTASK0010001 is created for the requested item
4	Task is assigned to the Hardware group	Flow (automatic)	Hardware team sees the task in its queue
5	Request is approved	Approver	Approval state changes to Approved
6	Laptop is configured	Hardware team	Task is worked on and closed

4.2 Flow Design

Table 4.2: Design of the flow "Standard Laptop task"

Element	Configuration
Flow name	Standard Laptop task
Application	Global
Run as	System User
Trigger	Service Catalog
Action	Create Catalog Task
Request item	Requested Item record from the trigger (drag and drop)
Table	Catalog Task [sc_task] - populated automatically
Short description	Laptop needs to Configured
Field: Description	Laptop needs to Configured
Field: Assignment group	Hardware
Field: Approval	Approved
Status	Saved and Activated

4.3 Flow Assignment Design

A flow does not run for a catalog item until it is attached to it. The Standard Laptop item was opened from **Maintain Items**, and on its **Process Engine** tab the flow "Standard Laptop task" was selected. Only one process engine can be used for an item, so the workflow and execution plan fields are left empty.

5. IMPLEMENTATION

The project was implemented in three milestones, all performed in the ServiceNow instance with the administrator account.

5.1 Milestone 1 - Creating the Flow

Activity 1: Create a flow to create a catalog task and assign it to the Hardware group

Creating the flow

1. Open ServiceNow.
2. Click **All** and search for **Flow Designer**.
3. Click **Flow Designer** under **Process Automation**.

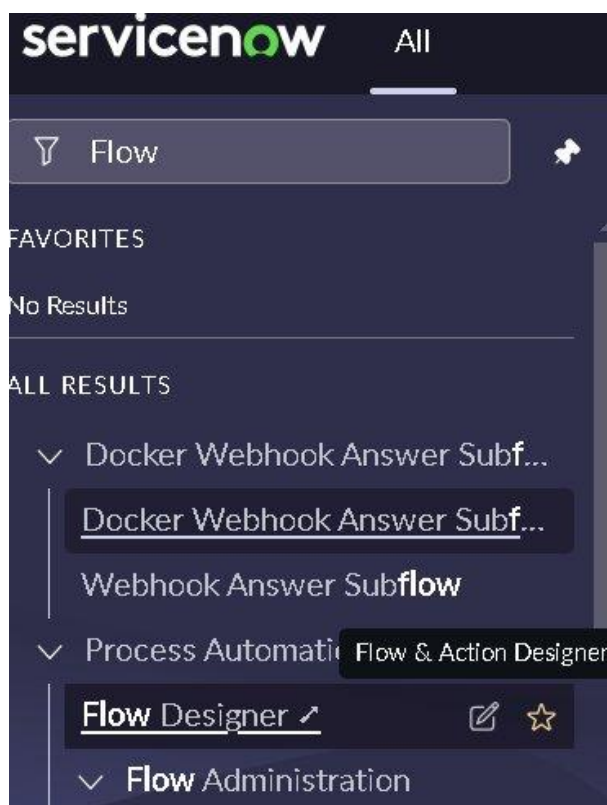


Figure 1: Searching for Flow Designer under Process Automation

4. In Flow Designer, click **New** and select **Flow**.

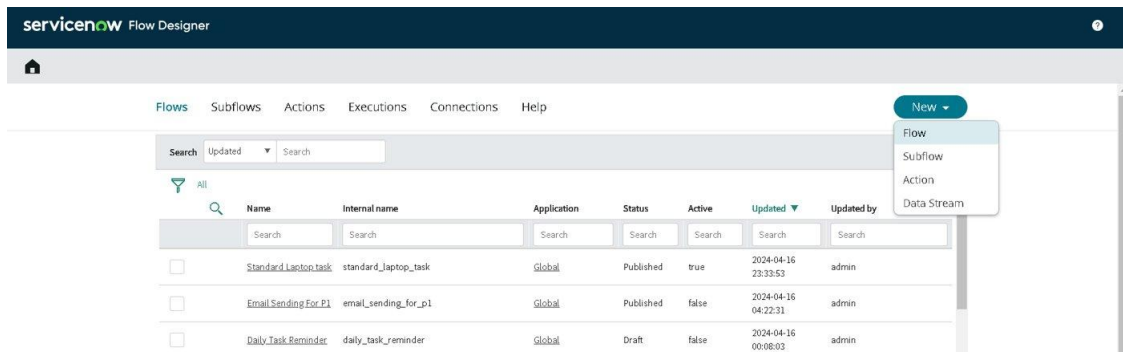


Figure 2: Flow Designer home page - New > Flow

5. In Flow properties, enter the Flow name **Standard Laptop task**.
6. Keep the Application as **Global**.
7. Select **System User** in the Run As field.
8. Click **Submit**.

Flow properties

* Flow name: Standard Laptop task

Description: Describe your flow

Application: Global

Protection: -- None --

Run As: System User

Cancel Submit

Figure 3: Flow properties dialog for "Standard Laptop task"

Adding the trigger

9. Click **Add a trigger**.
10. In the trigger list, search for **Service Catalog** and select it.
11. Click **Done**.

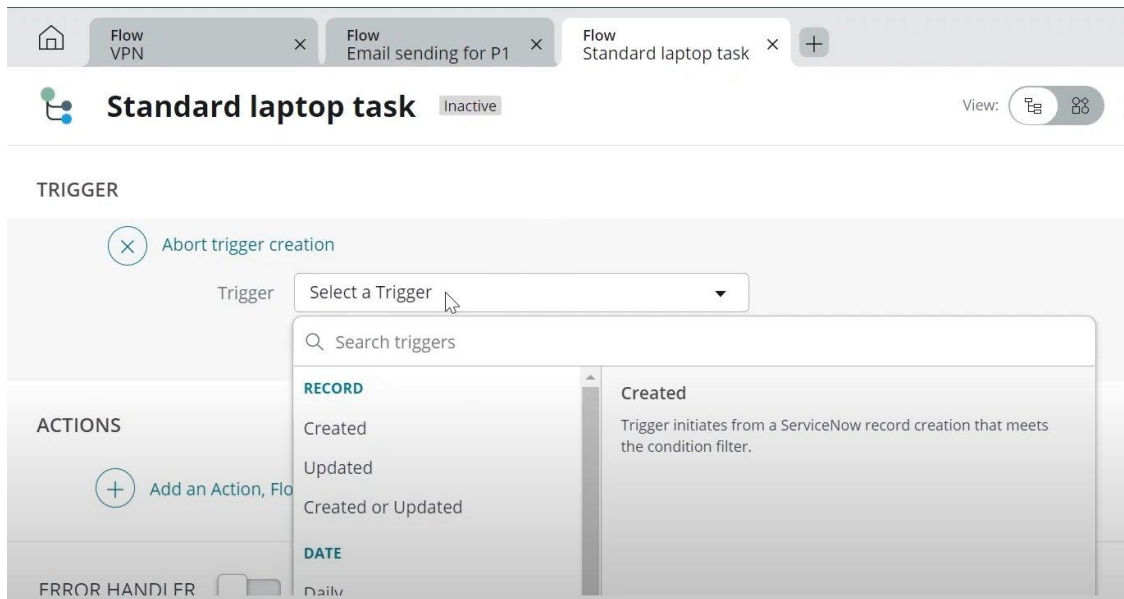


Figure 4: Selecting a trigger for the flow

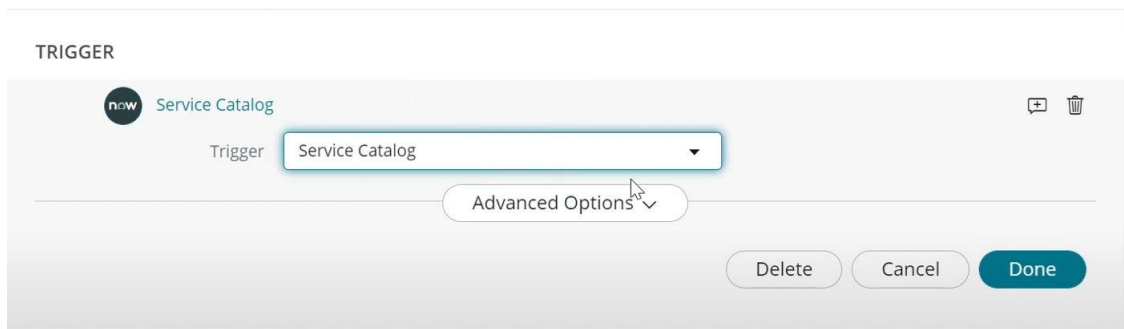


Figure 5: Service Catalog trigger added

Adding the action

12. Under Actions, click **Add an action**.
13. Search for **Create Catalog Task** and select it.
14. Set the Action to **Create Catalog Task**.
15. Drag and drop the **Requested Item** record from the trigger into the **Request Item** field.
16. The table is filled automatically as **Catalog Task [sc_task]**.
17. Enter the Short description **Laptop needs to Configured**.
18. Click **Add field value**. Select the **Description** field with the value **Laptop needs to Configured**.
19. Add the **Assignment group** field with the value **Hardware**.
20. Add the **Approval** field with the value **Approved**.
21. Leave the remaining fields as default and click **Done**.

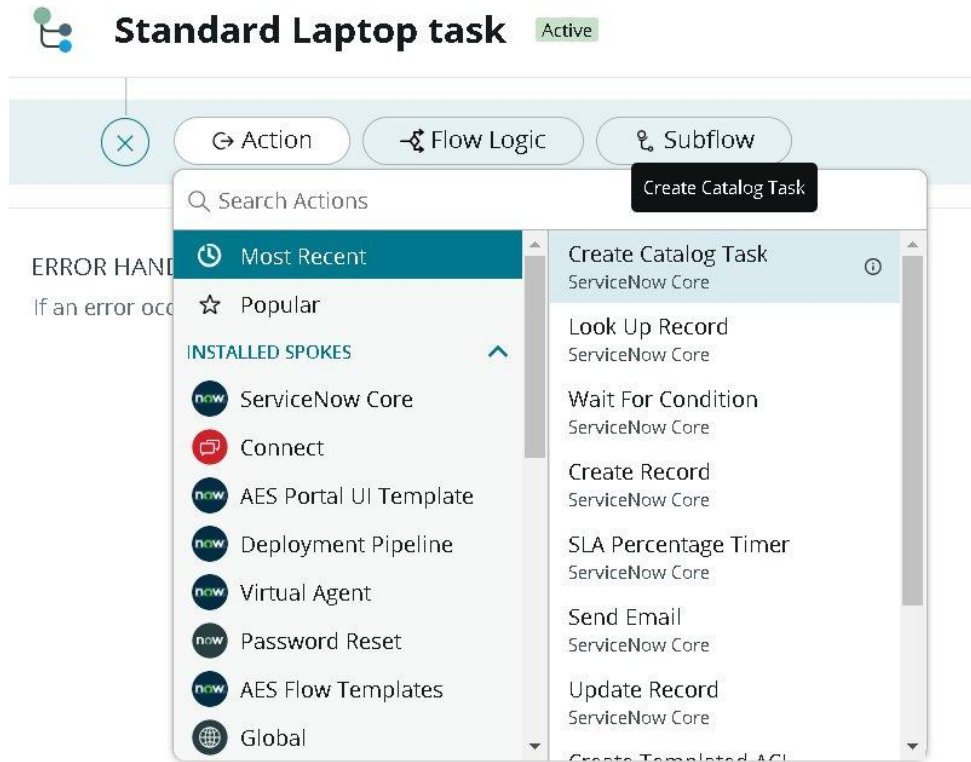


Figure 6: Searching for the Create Catalog Task action

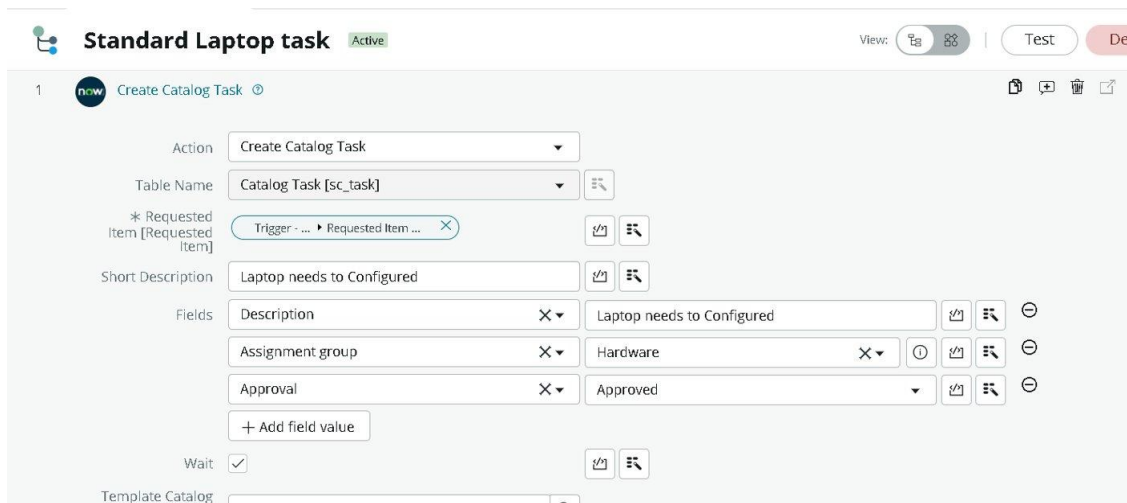


Figure 7: Create Catalog Task action configuration

Saving and activating the flow

22. Click **Save** to save the flow.
23. Click **Activate**, and click **Activate** again in the confirmation dialog.

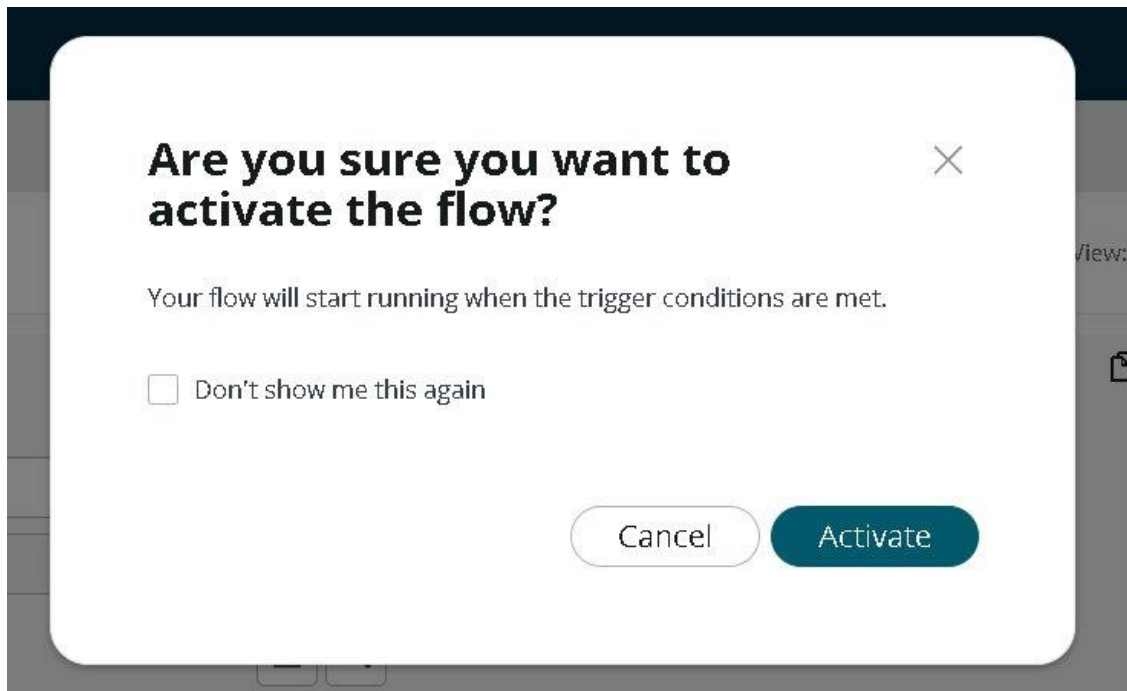


Figure 8: Confirmation dialog for activating the flow

5.2 Milestone 2 - Flow Assignment

Activity 1: Assign the flow to the Standard Laptop service catalog item

1. Open ServiceNow.
2. Click **All** and search for **Maintain Items**; select it.
3. In the Name column, search for **Standard Laptop** and open that record.

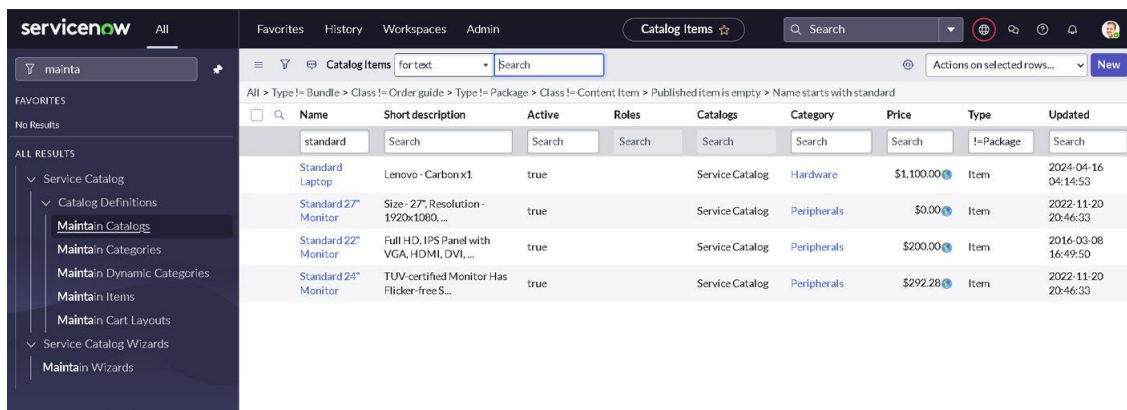


Figure 9: Locating the Standard Laptop item in Maintain Items

4. Open the **Process Engine** tab of the record.
5. Remove any existing automations and add the **Flow**.
6. Select the flow created earlier, **Standard Laptop task**.
7. Click the form menu and save the record (**Update**).

Catalog Item
Standard Laptop

Name: Standard Laptop

Application: Global

Catalogs: Service Catalog

Category: Hardware

State: -- None --

Checked out: -- None --

Owner:

Fulfillment automation level: Unspecified

Active: ☒

Update Copy Try It Edit in Catalog Builder De

Item Details **Process Engine** Picture Pricing Portal Settings

Select the appropriate process engine for the catalog Item. Only one engine can be selected.

Flow: Standard Laptop task

Workflow:

Execution Plan:

Update Copy Try It Edit in Catalog Builder Delete

Figure 10: Process Engine tab with the flow "Standard Laptop task" assigned

5.3 Milestone 3 - Service Catalog

Activity 1: Place the order and get approval

Placing the order

1. Click **All** and search for **Service Catalog**.
2. Open **Service Catalog** and click **Hardware**.
3. Scroll to find **Standard Laptop** and click it.
4. Click **Order Now** to place the request.

Favorites History Workspaces Admin

Standard Laptop

Search

Service Catalog > Hardware > Standard Laptop

Search catalog

Lenovo - Carbon x1

x1 Carbon

The x1 Carbon is Lenovo's lightest ThinkPad yet. It provides a QHD display that fights glare and weighs less than three pounds. Ideal for most computing tasks, and highly mobile.

Technical Specs:

- Intel core i5 processor
- 512GB solid state drive (SSD)
- Backlit keyboard

Optional Software

☐ Adobe Acrobat

☐ Adobe Photoshop

Additional software requirements

Order this Item

Price: ₹91,931.2306 + ₹8,357.3846 Annually

Quantity: 1

Subtotal: ₹91,931.2306 + ₹8,357.3846 Annually

Delivery: Order Now 0 Days

Order Now

Add to Cart

Shopping Cart: Empty

Figure 11: Standard Laptop catalog item with the Order Now button

5. The **Order Status** page is displayed.
6. Click the **Request Number** (REQ0010001).

Order Status

Back to Catalog
Continue Shopping
Home

Thank you, your request has been submitted

Order Placed: 2024-04-17 04:07:11
Request Number: REQ0010001 ☆
Estimated Delivery Date of Complete Order: 2024-04-17

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2024-04-17		₹91,931.2306 +₹8,357.3846 Annually	1	₹91,931.2306 +₹8,357.3846 Annually
				Total	₹91,931.2306 +₹8,357.3846 Annually

Back to Catalog
Continue Shopping

Home

Figure 12: Order Status page showing the request number REQ0010001

Approving the request

7. The request record opens. Scroll down and click the **Approvers** tab.

Favorites
History
Workspaces
Admin
Request - REQ0010001 ☆
Search

Request REQ0010001
Discuss
Follow
Update
Add New Item
Cancel Request
Copy
Delete

Short description
Special instructions

Update
Add New Item
Cancel Request
Copy
Delete

Related Links
[Show Workflow](#)
[Workflow Context](#)

Requested Items (1)
Approvers (1)
Recurring Prices (1)

State Search
Actions on selected rows... Edit

Approval for = REQ0010001

State	Approver	Comments	Created
Requested	Eric Schroeder		2024-04-17 04:07:12

1 to 1 of 1

Figure 13: Approvers tab of the request

8. Right-click the approval record and click **Approve**.

Update
Add New Item
Cancel Request
Copy
Delete

Related Links
[Show Workflow](#)
[Workflow Context](#)

Requested Items (1)
Approvers (1)
Recurring Prices (1)

State Search
Actions on selected rows... Edit

Approval for = REQ0010001

State	Approver	Comments	Created
Requested	Eric Schroeder		2024-04-17 04:07:12

1 to 1 of 1

Show Matching
Filter Out
Copy URL to Clipboard
Copy sys_id
Assign Tag
Approve
Reject

Figure 14: Approving the request from the context menu

Checking the catalog task created by the flow

9. Click the **Requested Items** tab and open the requested item record (RITM0010001).
10. Scroll down to the **Catalog Tasks** section, where the task created by the flow is listed with the Hardware assignment group.

The screenshot shows the 'Requested Item - RITM0010001' page. At the top, there's a navigation bar with 'Favorites', 'History', 'Workspaces', and 'Admin'. Below this, a search bar and a 'Search' button are visible. The main content area shows 'Activities: 1' with a list of activities for 'System Administrator'. The 'Catalog Tasks (1)' tab is selected, showing a table with one task: 'SCTASK0010001' assigned to the 'Hardware' group. The task's short description is 'Laptop needs to Configured' and its actual start date is '2024-04-17 04:11:40'.

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010001	Hardware	(empty)	Laptop needs to Configured	2024-04-17 04:11:40	(empty)	1

Figure 15: Requested item RITM0010001 with the auto-created catalog task

11. Open the catalog task (SCTASK0010001) to see the updated short description and assignment group.

The screenshot shows the 'Catalog Task - SCTASK0010001' page. The task details are displayed, including the 'Number' (SCTASK0010001), 'Assigned to' (empty), 'Configuration item' (empty), 'Active' (checked), 'Approval' (Approved), 'Priority' (4 - Low), 'State' (Open), 'Request item' (RITM0010001), and 'Requested for' (System Administrator). The 'Short description' and 'Description' fields both contain 'Laptop needs to Configured'. The 'Worknotes' field is empty. A 'Post' button is visible at the bottom right.

Figure 16: Catalog task SCTASK0010001 with short description and Hardware group

6. TESTING AND RESULTS

6.1 Test Cases

Table 6.1: Test cases and observed results

S.No	Test	Expected result	Observed result
1	Flow activated	Status shows Active	Flow active - Figures 7, 8
2	Flow attached to Standard Laptop item	Process Engine shows the flow	Flow "Standard Laptop task" selected - Figure 10
3	Order Standard Laptop from the catalog	Request is created	REQ0010001 created - Figure 12
4	Approve the request	Approval state becomes Approved	Approval completed - Figure 14
5	Check the Requested Item	A catalog task is listed	SCTASK0010001 listed - Figure 15
6	Check the catalog task	Short description and group are set	Laptop needs to Configured; group Hardware - Figure 16

6.2 Result Analysis

After the order was placed and approved, the catalog task SCTASK0010001 was present under the requested item RITM0010001 without any manual creation. Its short description read "Laptop needs to Configured", the assignment group was Hardware and the approval value was Approved, exactly as configured in the flow. This confirms that the trigger, the action and the catalog item assignment work together correctly.

7. ADVANTAGES AND LIMITATIONS

7.1 Advantages

- Removes the manual step of creating configuration tasks, so nothing is forgotten.
- Reduces the time a user waits for a configured laptop.
- Tasks always reach the correct team, improving use of IT staff.
- Built with low-code configuration, so it is easy to understand and change.
- The same design can be reused for other catalog items.

7.2 Limitations

- The flow is attached to one catalog item only (Standard Laptop).
- Short description, assignment group and approval value are fixed in the action rather than read from item variables.
- The flow does not send e-mail notifications to the requester or the hardware team.
- Vendor ordering and asset records are not automated.

8. CONCLUSION

By automating the standard laptop procurement process with ServiceNow's Flow Designer, the project effectively addresses inefficiencies and reduces manual overhead. The workflow ensures timely configuration of laptops, minimizes user wait times and enhances overall productivity within the IT department. The automated solution also optimizes resource allocation and improves the user experience by providing a seamless and efficient procurement process.

Through this work, I gained practical experience with Flow Designer, triggers and actions, the Service Catalog, catalog items, approvals and catalog tasks. These are core skills for a ServiceNow developer or administrator and have strengthened my readiness for entry-level roles in the IT industry.

9. FUTURE SCOPE

- Send e-mail or notification messages to the requester and to the Hardware group when the task is created and closed.
- Use catalog item variables (such as optional software) to add more details to the task automatically.
- Create the task only after approval using flow logic such as "Ask for Approval" and conditions.
- Add a follow-up task for software installation and a final task to deliver the laptop.
- Create asset and configuration item records automatically once the laptop is configured.
- Extend the flow to other catalog items such as monitors and peripherals.

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<https://developer.servicenow.com>
4. Naan Mudhalvan (TNSkills) - ServiceNow course material.
<https://naanmudhalvan.tn.gov.in>

APPENDIX A - FLOW CONFIGURATION SUMMARY

Table A.1: Create Catalog Task action settings

Field	Value
Action	Create Catalog Task
Table name	Catalog Task [sc_task]
Requested Item	Trigger - Service Catalog > Requested Item Record
Short description	Laptop needs to Configured
Description	Laptop needs to Configured
Assignment group	Hardware
Approval	Approved
Wait	Selected (default)

APPENDIX B - RECORDS CREATED DURING TESTING

Table B.1: Records generated during the test

Record	Value
Catalog item ordered	Standard Laptop (Lenovo - Carbon x1: Intel Core i5, 512 GB SSD, backlit keyboard)
Request number	REQ0010001
Requested item	RITM0010001
Catalog task	SCTASK0010001
Approver	Eric Schroeder
Assignment group	Hardware